Lab_sheet-03

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Section→C

Roll no.→11

Experiment No. 3

Experiment Name

Implementation and Analysis of Spatial Filtering Techniques using Low-Pass and High-Pass Filters

Program Code and Output

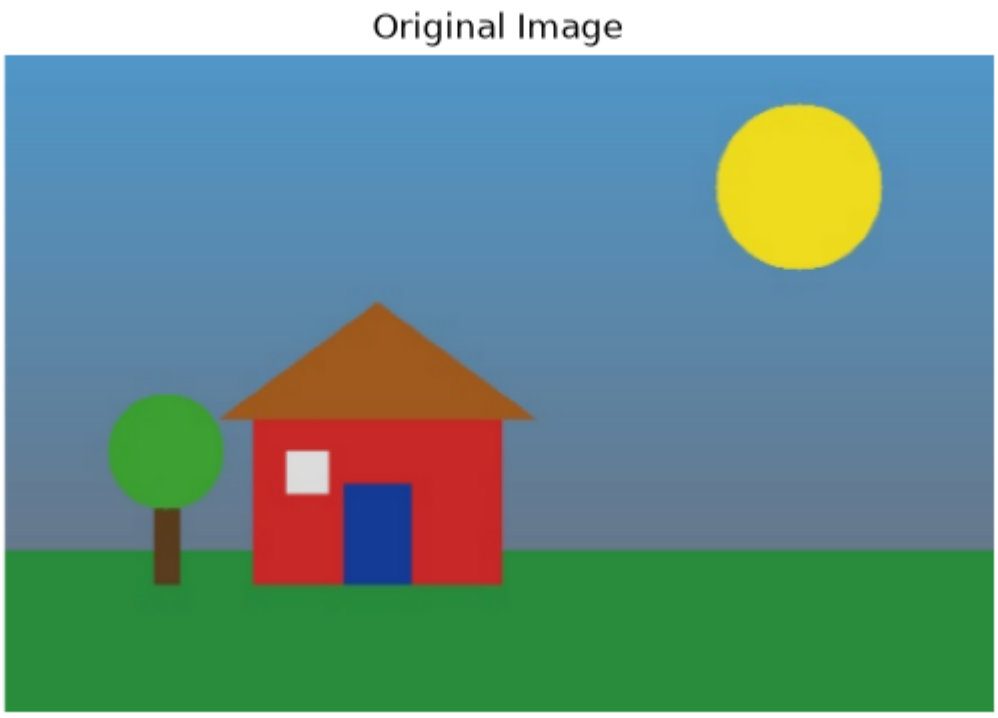
Step 1–2: Libraries import + image load/display

In [1]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load image
image = cv2.imread("my_pic01.png")

# Convert BGR to RGB for displaying
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Display original image
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")
plt.show()
```



Step 3: Gaussian Blur

In [2]:
```
gaussian = cv2.GaussianBlur(image, (5, 5), 0)

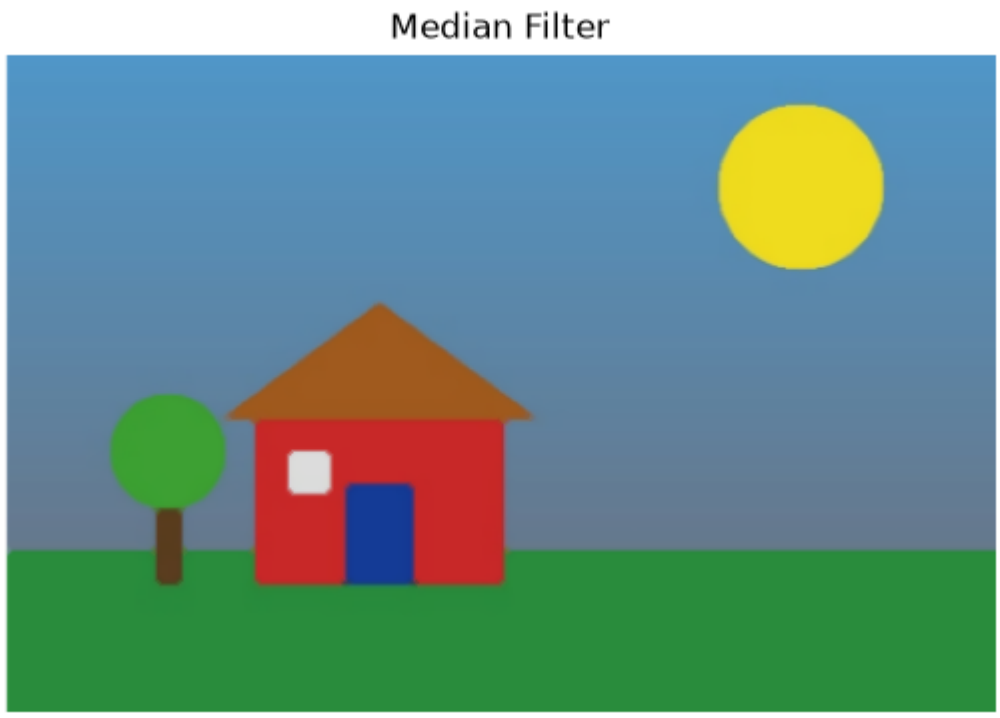
plt.imshow(cv2.cvtColor(gaussian, cv2.COLOR_BGR2RGB))
plt.title("Gaussian Blur")
plt.axis("off")
plt.show()
```



Step 4: Median Filtering

In [3]:
```
median = cv2.medianBlur(image, 5)

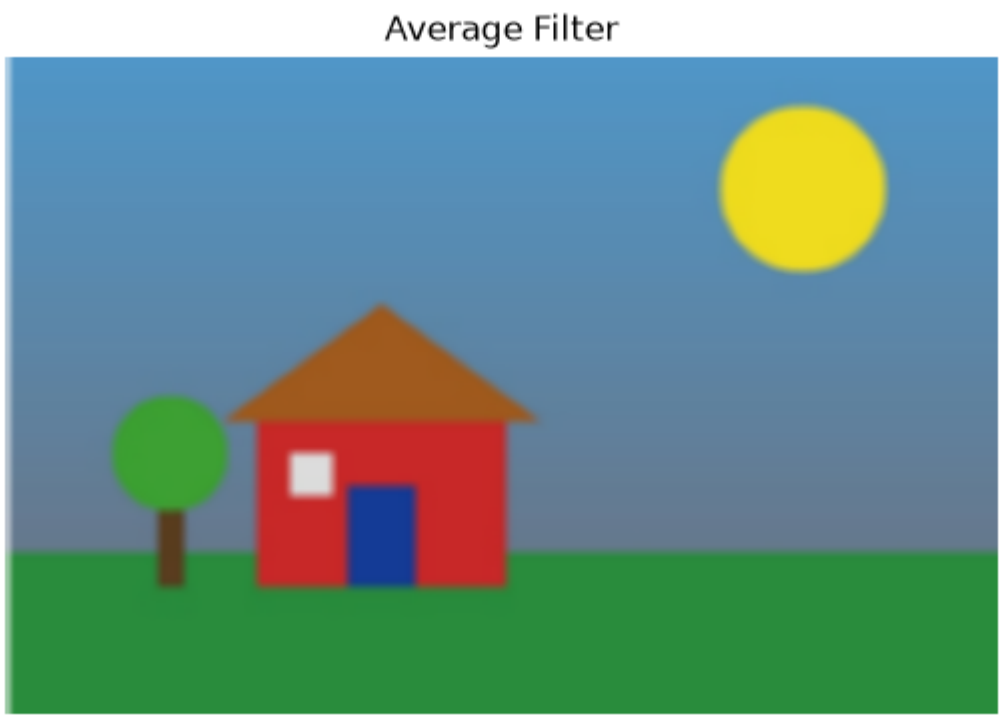
plt.imshow(cv2.cvtColor(median, cv2.COLOR_BGR2RGB))
plt.title("Median Filter")
plt.axis("off")
plt.show()
```



Step 5: Average (Mean) Filtering

In [4]:
```
average = cv2.blur(image, (5, 5))

plt.imshow(cv2.cvtColor(average, cv2.COLOR_BGR2RGB))
plt.title("Average Filter")
plt.axis("off")
plt.show()
```

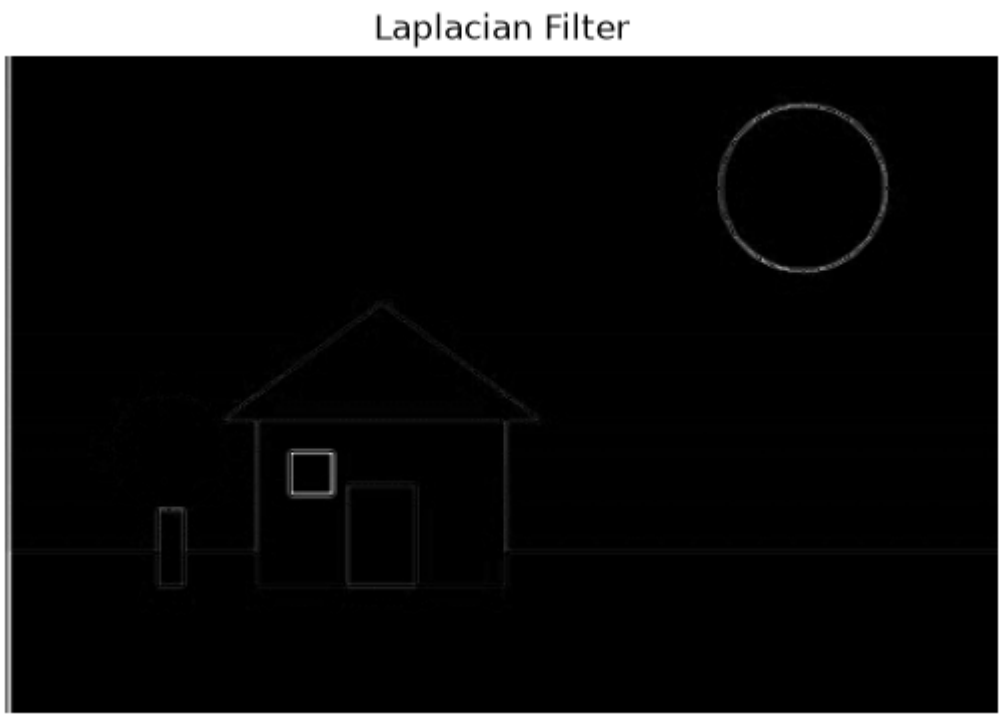


Step 6: Laplacian Filtering

In [5]:
```
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

laplacian = cv2.Laplacian(gray, cv2.CV_64F)

plt.imshow(np.abs(laplacian), cmap="gray")
plt.title("Laplacian Filter")
plt.axis("off")
plt.show()
```



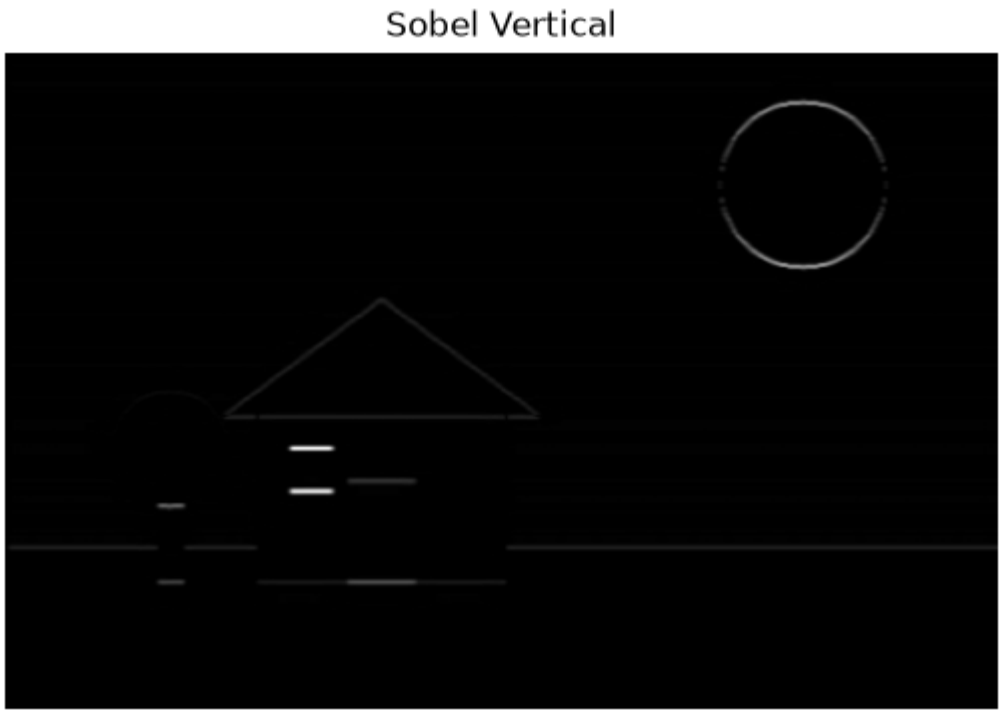
Step 7: Sobel Edge Detection — Horizontal & Vertical

In [6]:
```
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

plt.imshow(np.abs(sobel_x), cmap="gray")
plt.title("Sobel Horizontal")
plt.axis("off")
plt.show()

plt.imshow(np.abs(sobel_y), cmap="gray")
plt.title("Sobel Vertical")
plt.axis("off")
plt.show()
```





Step 8: Compare all filters

In [7]:
```
plt.figure(figsize=(12, 8))

plt.subplot(2, 3, 1)
plt.imshow(image_rgb)
plt.title("Original")
plt.axis("off")

plt.subplot(2, 3, 2)
plt.imshow(cv2.cvtColor(gaussian, cv2.COLOR_BGR2RGB))
plt.title("Gaussian")
plt.axis("off")

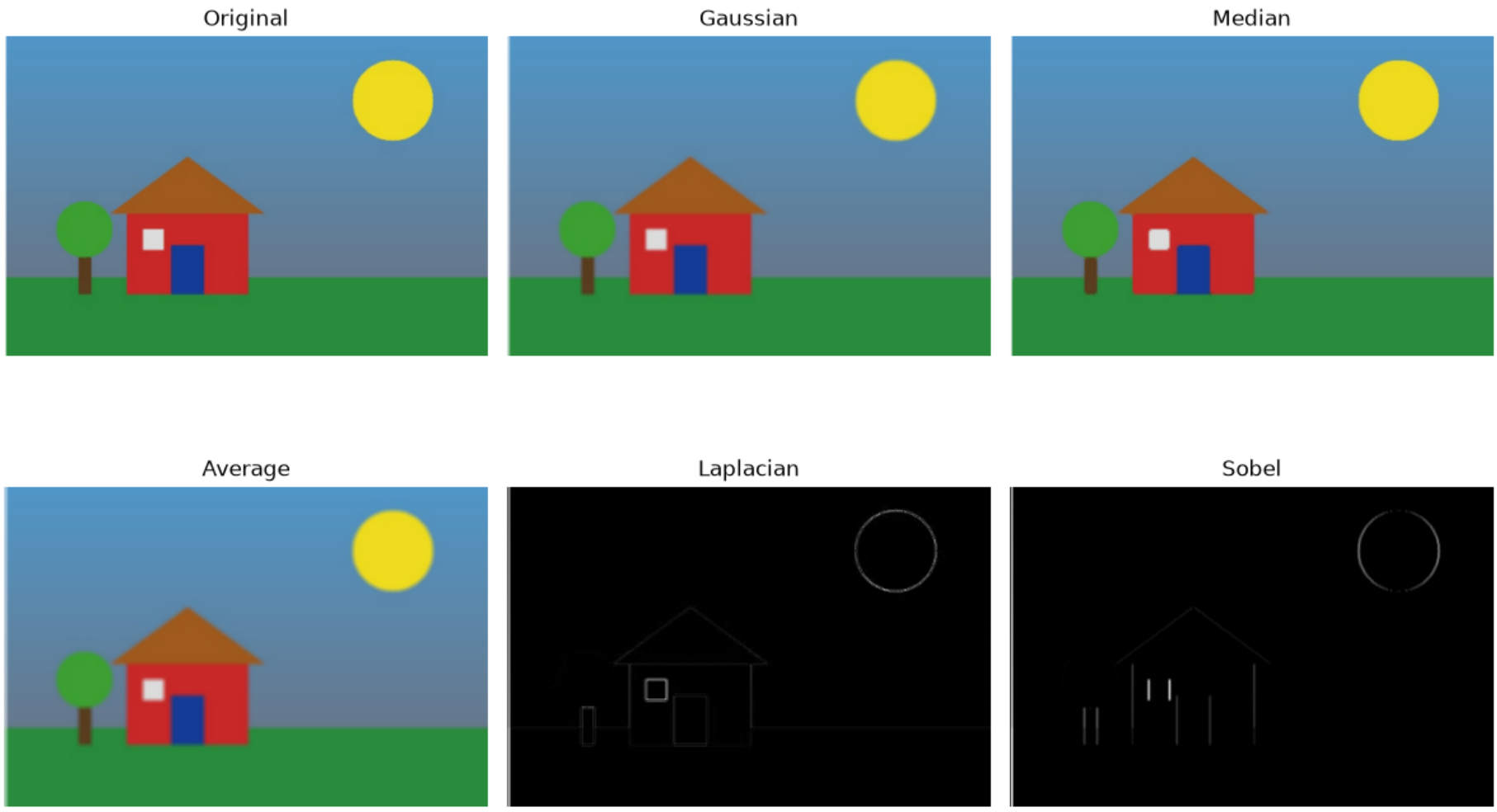
plt.subplot(2, 3, 3)
plt.imshow(cv2.cvtColor(median, cv2.COLOR_BGR2RGB))
plt.title("Median")
plt.axis("off")

plt.subplot(2, 3, 4)
plt.imshow(cv2.cvtColor(average, cv2.COLOR_BGR2RGB))
plt.title("Average")
plt.axis("off")

plt.subplot(2, 3, 5)
plt.imshow(np.abs(laplacian), cmap="gray")
plt.title("Laplacian")
plt.axis("off")

plt.subplot(2, 3, 6)
plt.imshow(np.abs(sobel_x), cmap="gray")
plt.title("Sobel")
plt.axis("off")

plt.tight_layout()
plt.show()
```



Step 9: Observations

In [8]:
```
print("""Gaussian Filter:
Reduces noise and produces smooth results while preserving important structures.

Median Filter:
Effective for removing salt-and-pepper noise and preserves edges comparatively well.

Average Filter:
Provides simple smoothing but may blur edges.

Laplacian Filter:
Enhances edges and fine details.

Sobel Filter:
Detects edges in horizontal and vertical directions.""")
```

```
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```

Step 10: Interpretation

Spatial filtering is useful for improving image quality before further computer vision operations. Low-pass filters such as Gaussian, Median, and Average filters are mainly used for smoothing and noise reduction. High-pass filters such as Sobel and Laplacian are useful for detecting edges and enhancing fine details.